

James Lewis - 29

Assignment 1:-

Create a simple program to manage products in a store using Python. Each product will have a name, price, and stock quantity. The program should allow adding products, updating stock, and viewing product details.

Manage products in a store using Python.

Define Product Class

```
In [1]: class Product:
        def __init__(self, name, price, stock):
            self.name = name
            self.price = price
            self.stock = stock

        def update_stock(self, quantity):
            self.stock += quantity
            print("Stock updated successfully!")

        def __str__(self):
            return f"Product: {self.name}, Price: ₹{self.price:.2f}, Stock: {self.stock}"
```

Initialize Product Dictionary

```
In [2]: products = {}
```

Function to Add Product

```
In [3]: def add_product():
        name = input("Enter product name: ")
        price = float(input("Enter product price: "))
        stock = int(input("Enter product stock: "))
        products[name] = Product(name, price, stock)
        print("Product added successfully!")
```

Function to Update Stock

```
In [4]: def update_product():
        name = input("Enter product name to update: ")
        if name in products:
            quantity = int(input("Enter quantity to add/remove: "))
```

```
        products[name].update_stock(quantity)
    else:
        print("Product not found!")
```

Function to View Product Details

```
In [5]: def view_product():
        name = input("Enter product name: ")
        if name in products:
            print(products[name])
        else:
            print("Product not found!")
```

Interactive Menu for User

```
In [6]: while True:
        print("\n1. Add Product\n2. Update Stock\n3. View Product\n4. Exit")
        choice = input("Enter your choice: ")

        if choice == '1':
            add_product()
        elif choice == '2':
            update_product()
        elif choice == '3':
            view_product()
        elif choice == '4':
            print("Exiting the system. Goodbye!")
            break
        else:
            print("Invalid choice. Please enter a number between 1 and 4.")
```

```
1. Add Product
2. Update Stock
3. View Product
4. Exit
Enter your choice: 1
Enter product name: BIKE
Enter product price: 250000
Enter product stock: 10
Product added successfully!
```

```
1. Add Product
2. Update Stock
3. View Product
4. Exit
Enter your choice: CAR
Invalid choice. Please enter a number between 1 and 4.
```

```
1. Add Product
2. Update Stock
3. View Product
4. Exit
Enter your choice: 1
Enter product name: CAR
Enter product price: 1000000
Enter product stock: 5
Product added successfully!
```

```
1. Add Product
2. Update Stock
3. View Product
4. Exit
Enter your choice: 3
Enter product name: BIKE
Product: BIKE, Price: ₹250000.00, Stock: 10
```

```
1. Add Product
2. Update Stock
3. View Product
4. Exit
Enter your choice: 4
Exiting the system. Goodbye!
```